

Math Hallucination Detector: Detecting and Explaining Mathematical Hallucinations in Large Language Models

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Abstract

Large language models often produce answers that appear mathematically plausible but are incorrect, creating reliability challenges in educational, scientific, and financial contexts. The **Math Hallucination Detector** automatically identifies and explains these errors.

The system verifies model outputs against ground-truth solutions from Wolfram Alpha and applies deterministic numeric and symbolic comparison rules. It also uses first-order predicate logic to generate structured reasoning traces and formal correctness checks.

For equation-based claims, the framework performs step-by-step expression evaluation and backward error analysis to locate where reasoning diverges. The system outputs clear verdicts, confidence scores, and interpretable explanations, providing a transparent method for detecting mathematical hallucinations in large language models.

Why It Matters

Educational reliability: Explains where reasoning fails, helping students learn instead of simply flagging wrong answers.

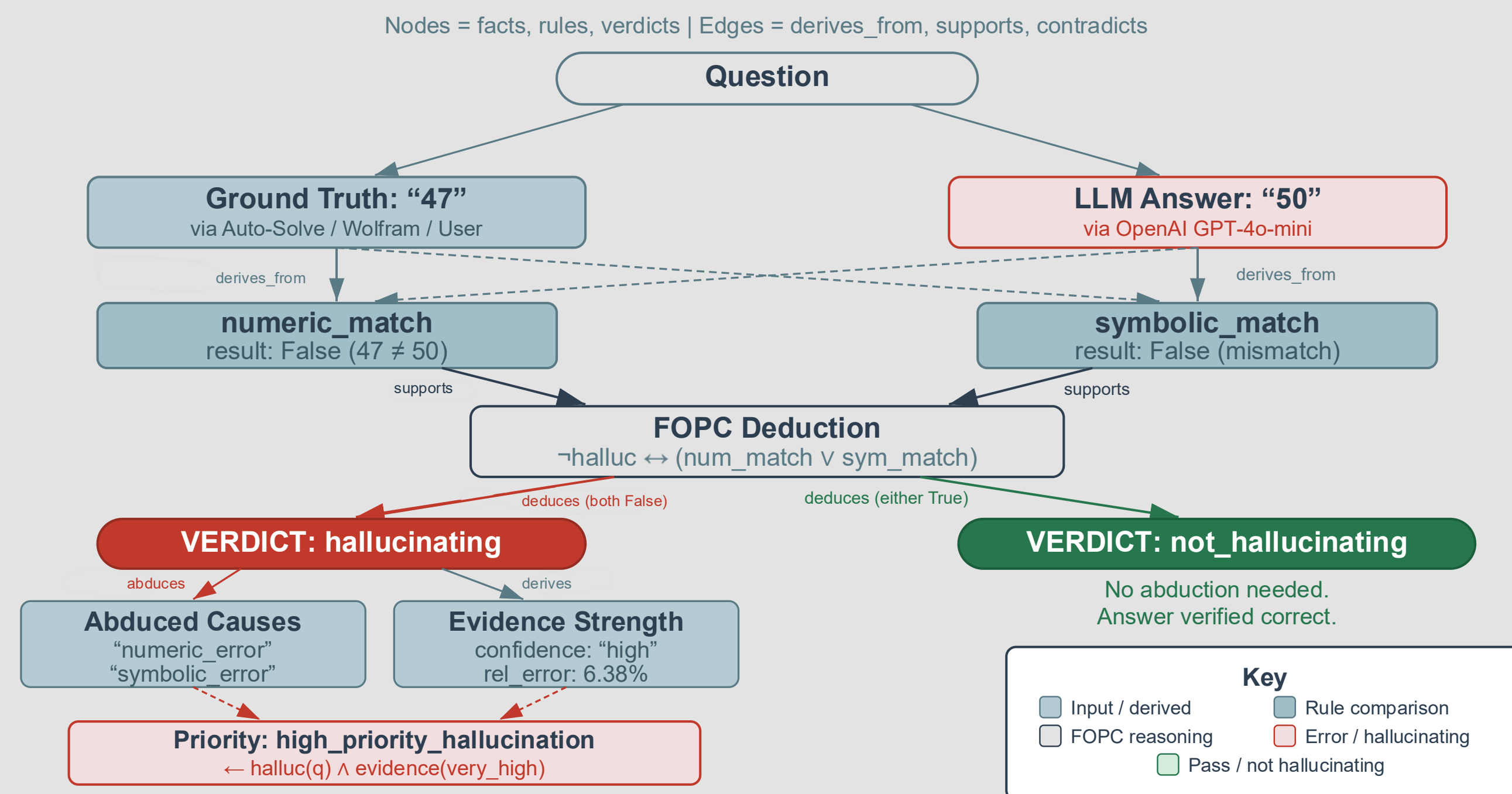
Trust in AI systems: Detecting mathematical hallucinations improves confidence in AI-generated answers used for learning and research.

Transparency and accountability: Produces auditable reasoning traces and deterministic verdicts, making outputs easier to verify and debug.

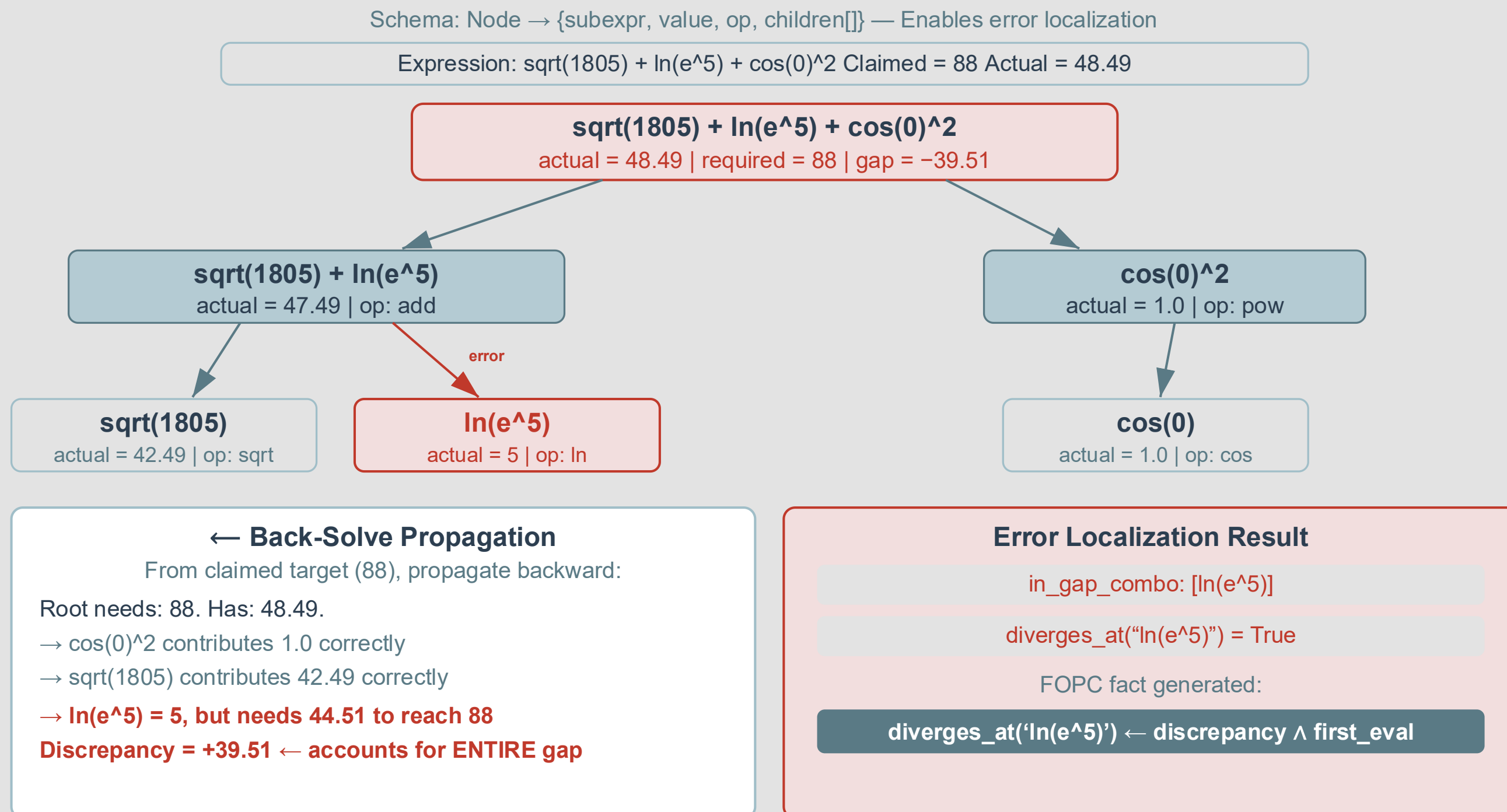
Scalable verification: Automatically generates and validates ground truth, enabling large-scale evaluation with minimal human review.

Model evaluation and improvement: Reveals patterns in hallucinations, helping researchers improve training methods and evaluation benchmarks.

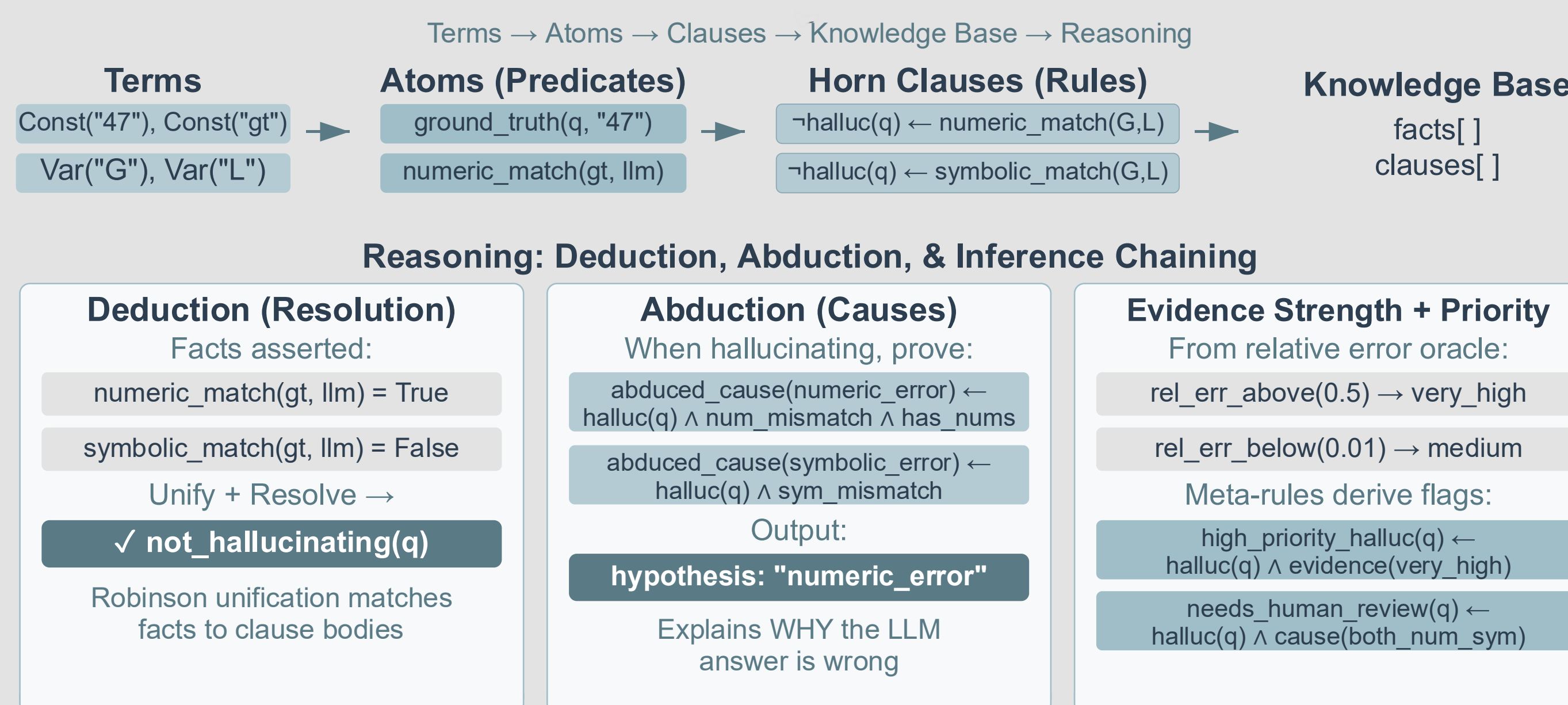
Reasoning Trace as Knowledge Graph



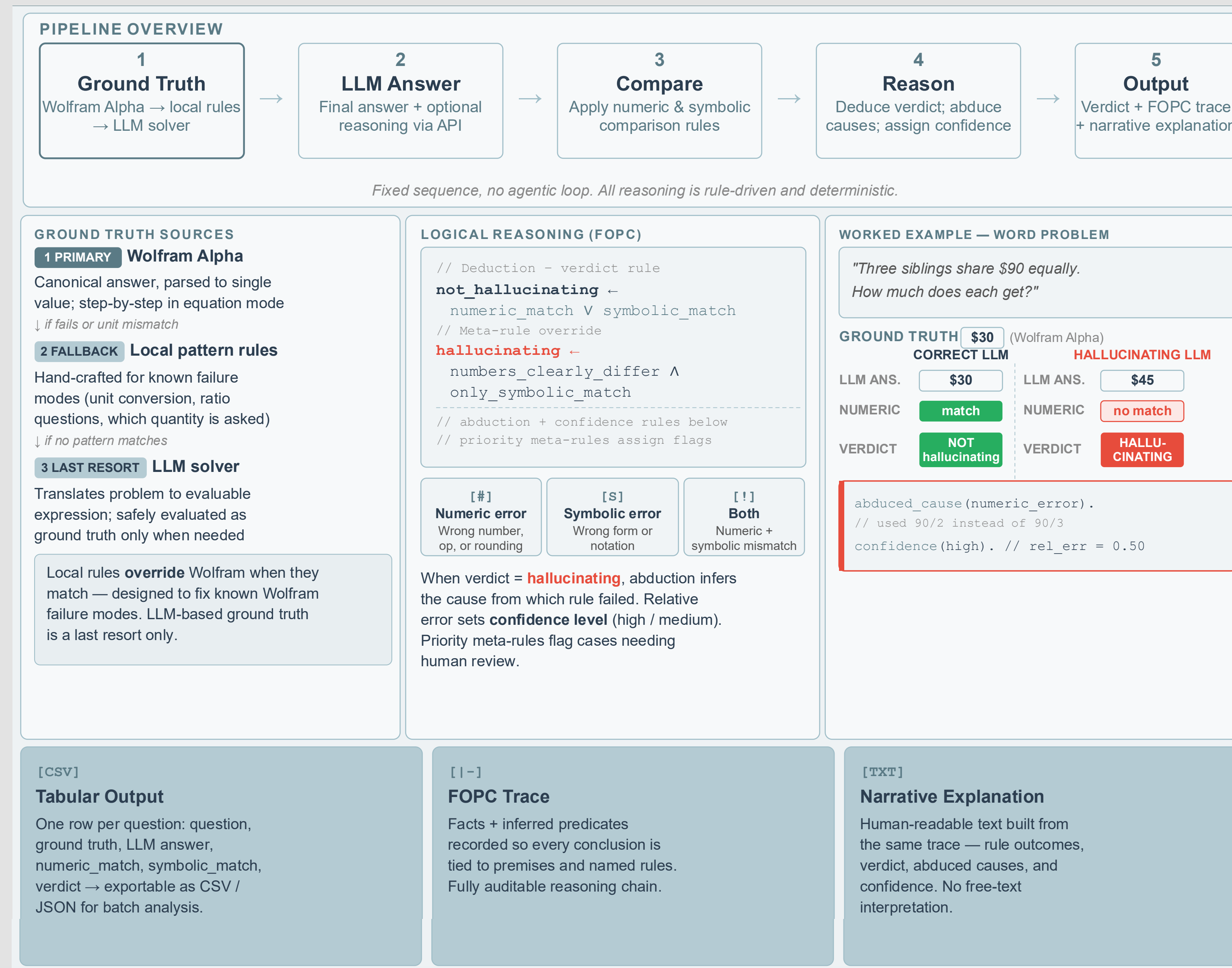
Reasoning Trace as Expression Tree



FOPC Knowledge Base Schema



Methods and How Reasoning Works



Conclusion

The **Math Hallucination Detector** shows that combining external computational verification with rule based logical reasoning offers a reliable way to detect mathematical hallucinations in LLM outputs. By validating answers against authoritative ground truth and applying deterministic numeric and symbolic comparison rules, the system can automatically identify discrepancies between generated and correct solutions.

Beyond simple answer checking, the framework produces structured reasoning traces, expression level evaluations, and FOPC based deductions that localize where reasoning diverges and infer likely causes of error. This enables transparent, auditable verification rather than opaque correctness judgments.

Overall, the project demonstrates that integrating symbolic reasoning, formal logic, and external computation can significantly improve reliability and interpretability of AI generated mathematical reasoning.

Future Work: Extending the framework to support broader mathematical domains and large-scale benchmark evaluation of LLM reasoning.